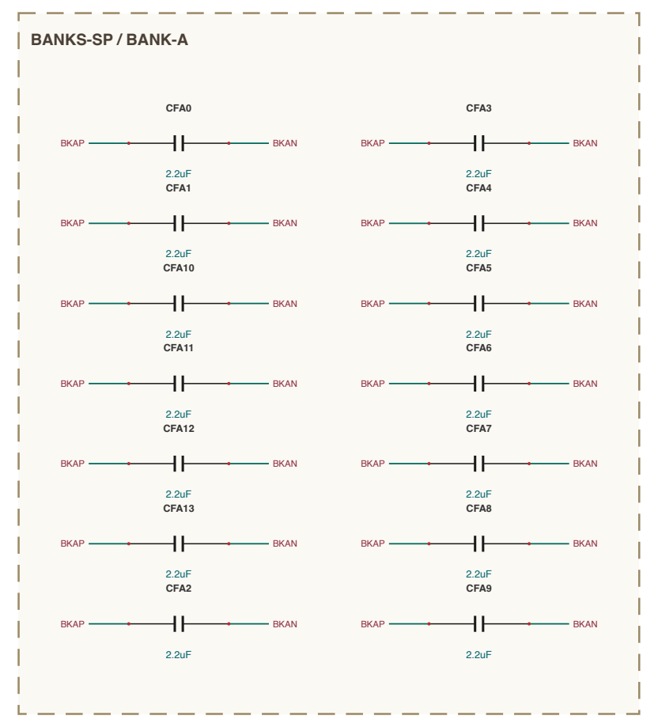

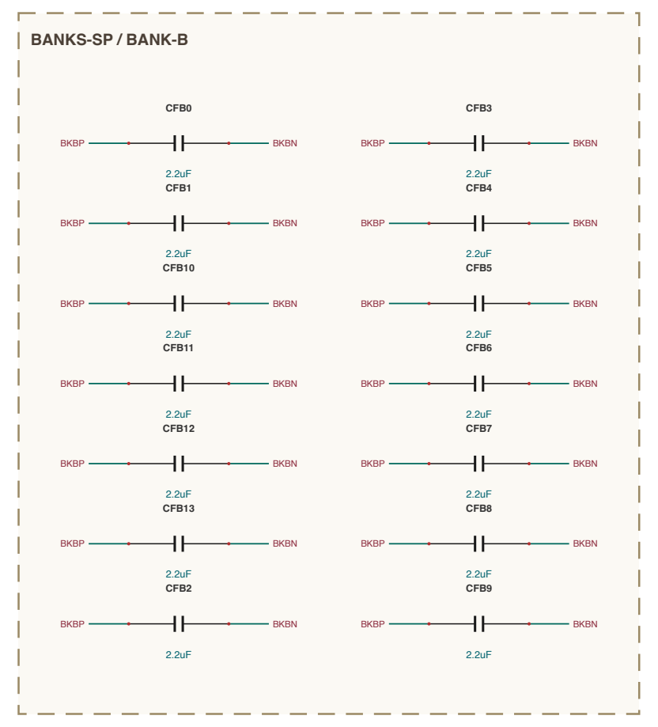

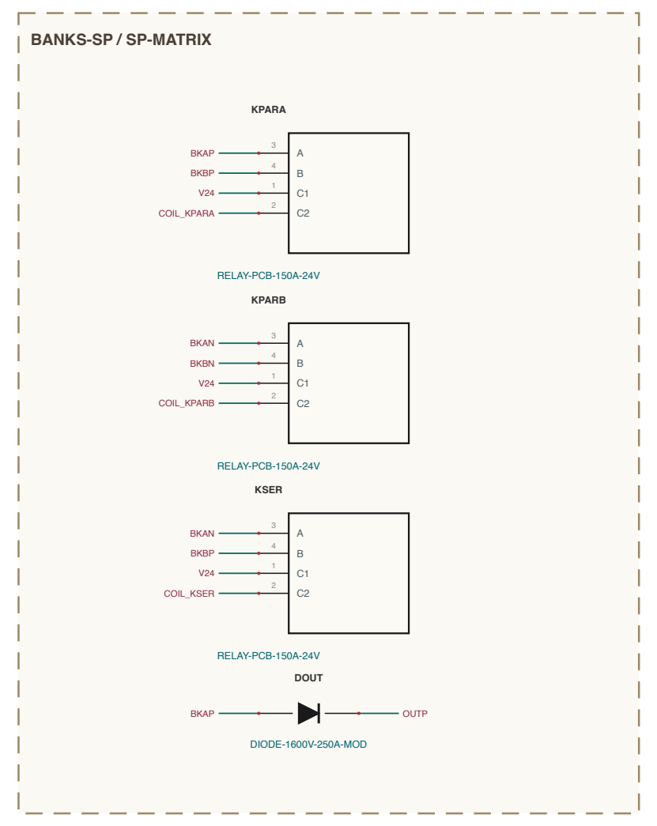

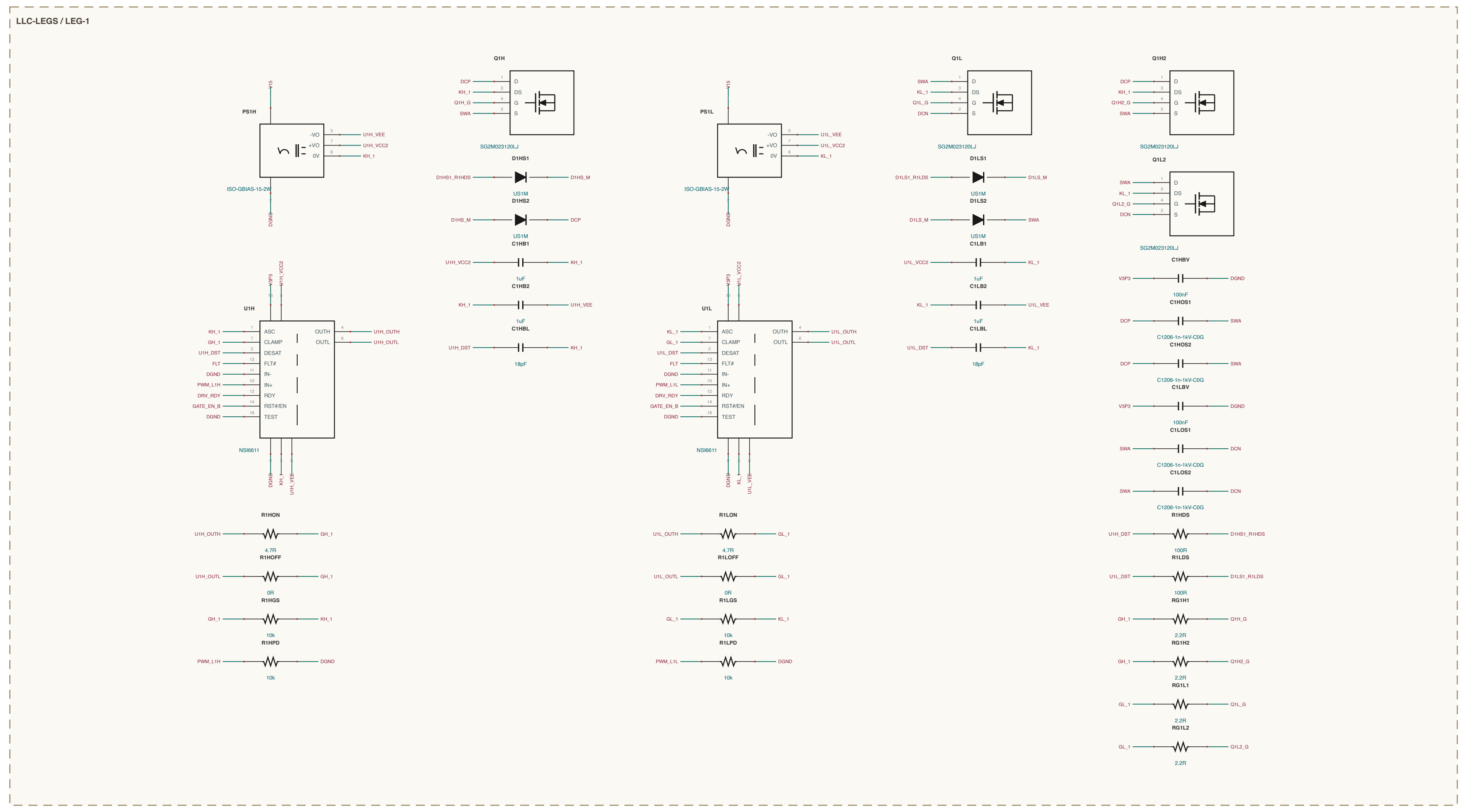

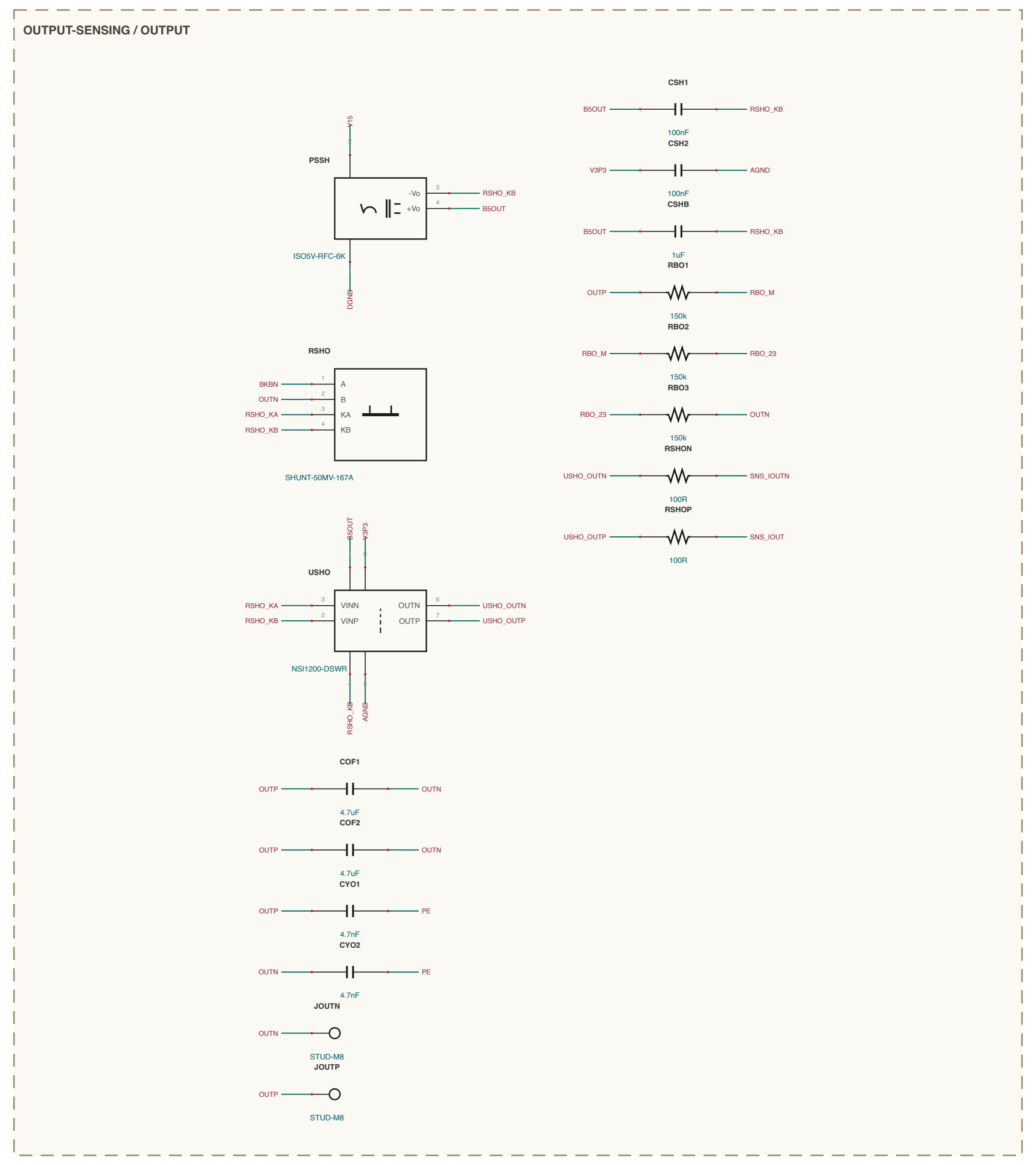

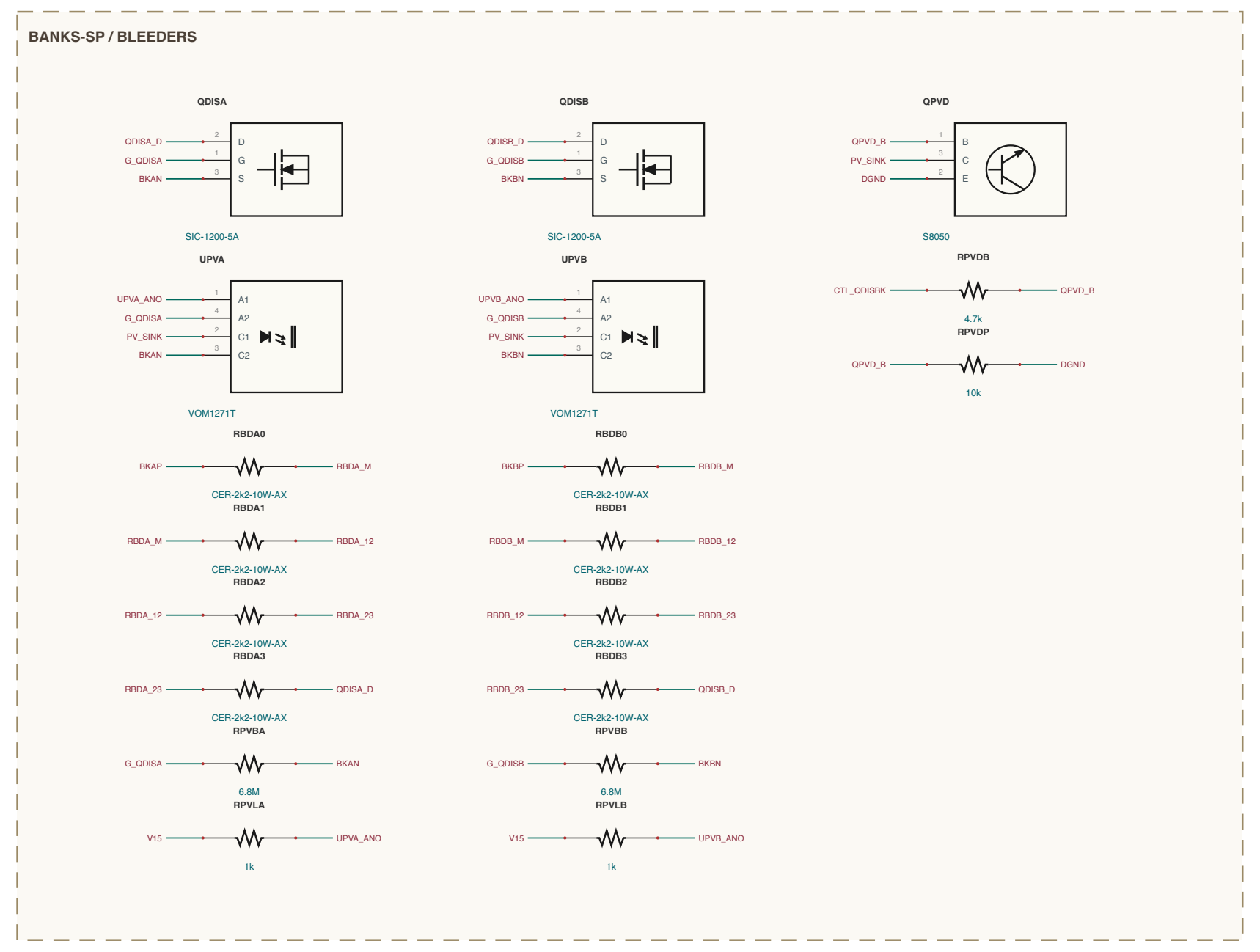

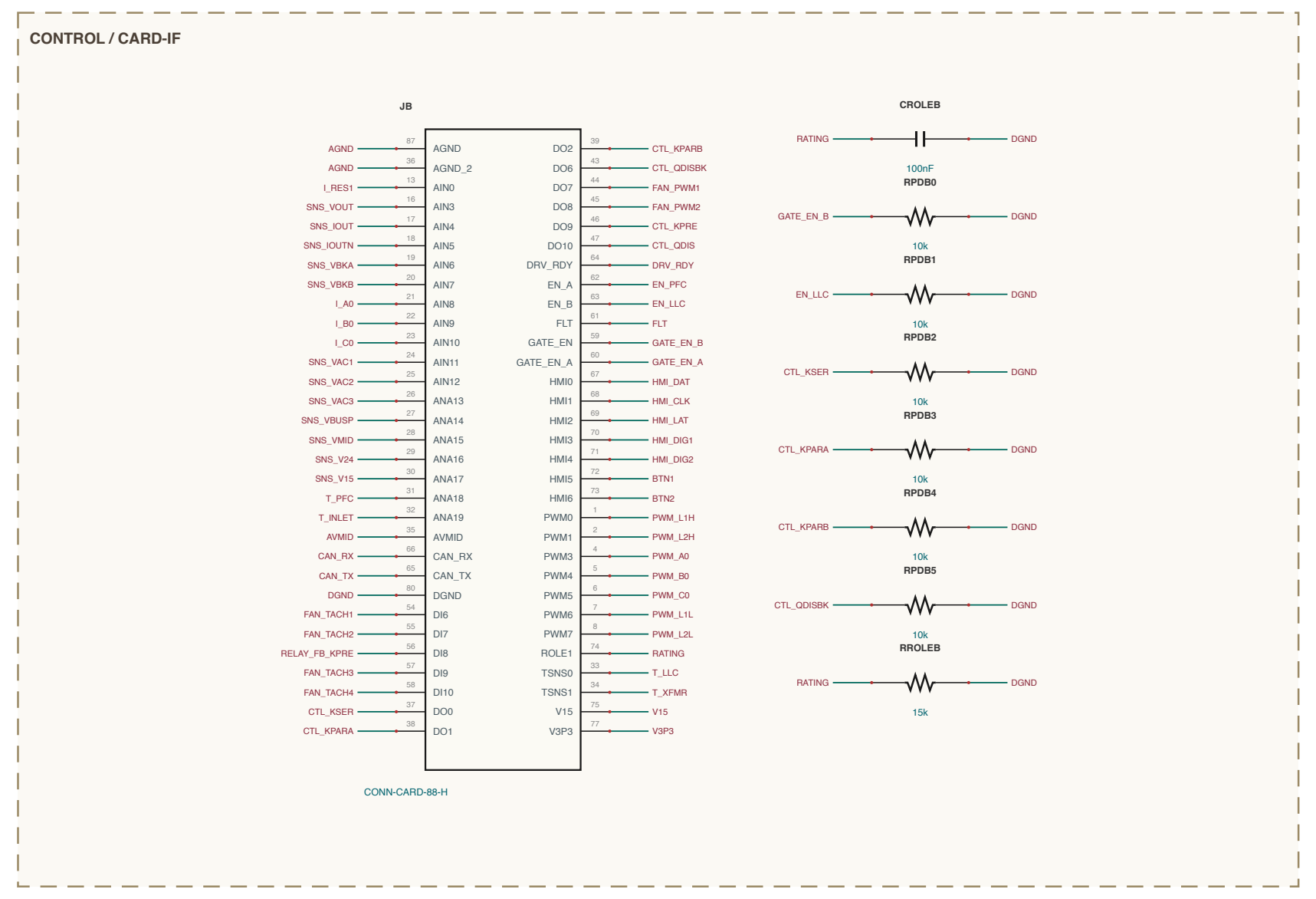

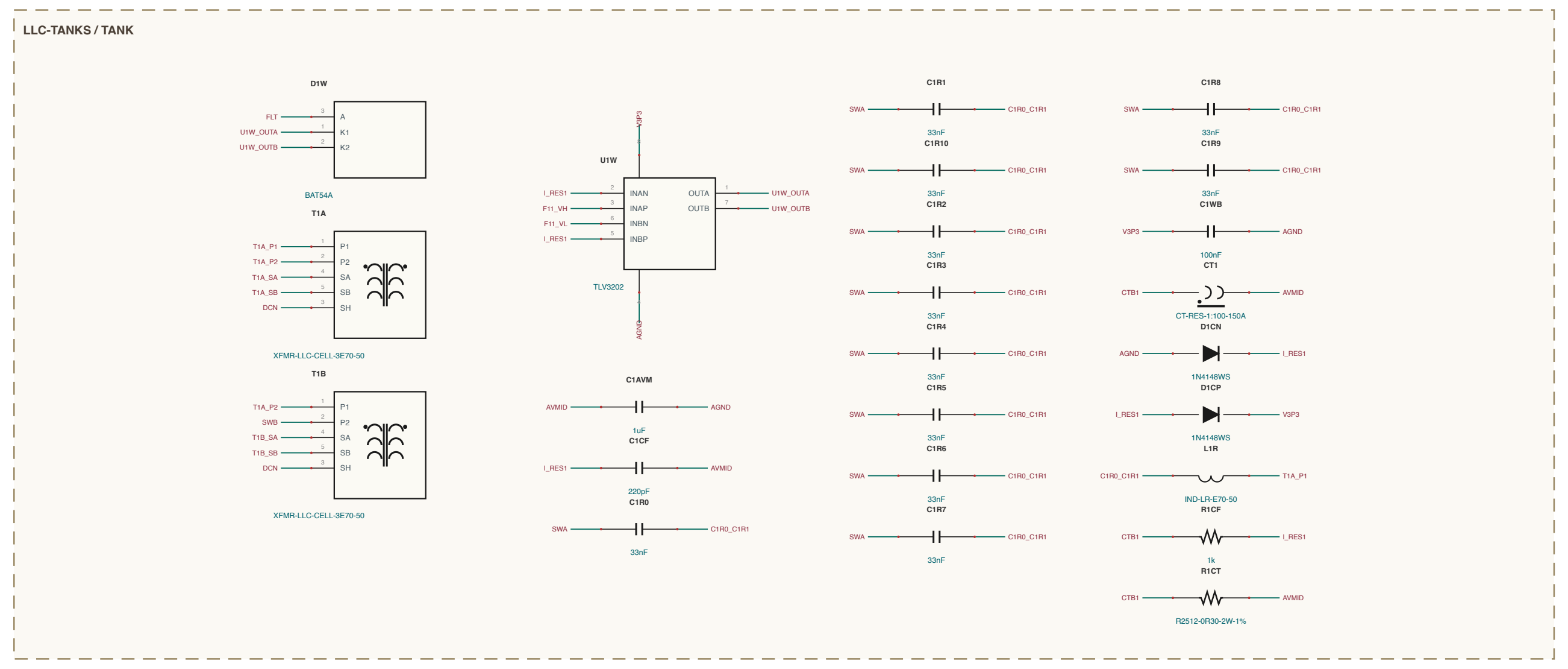

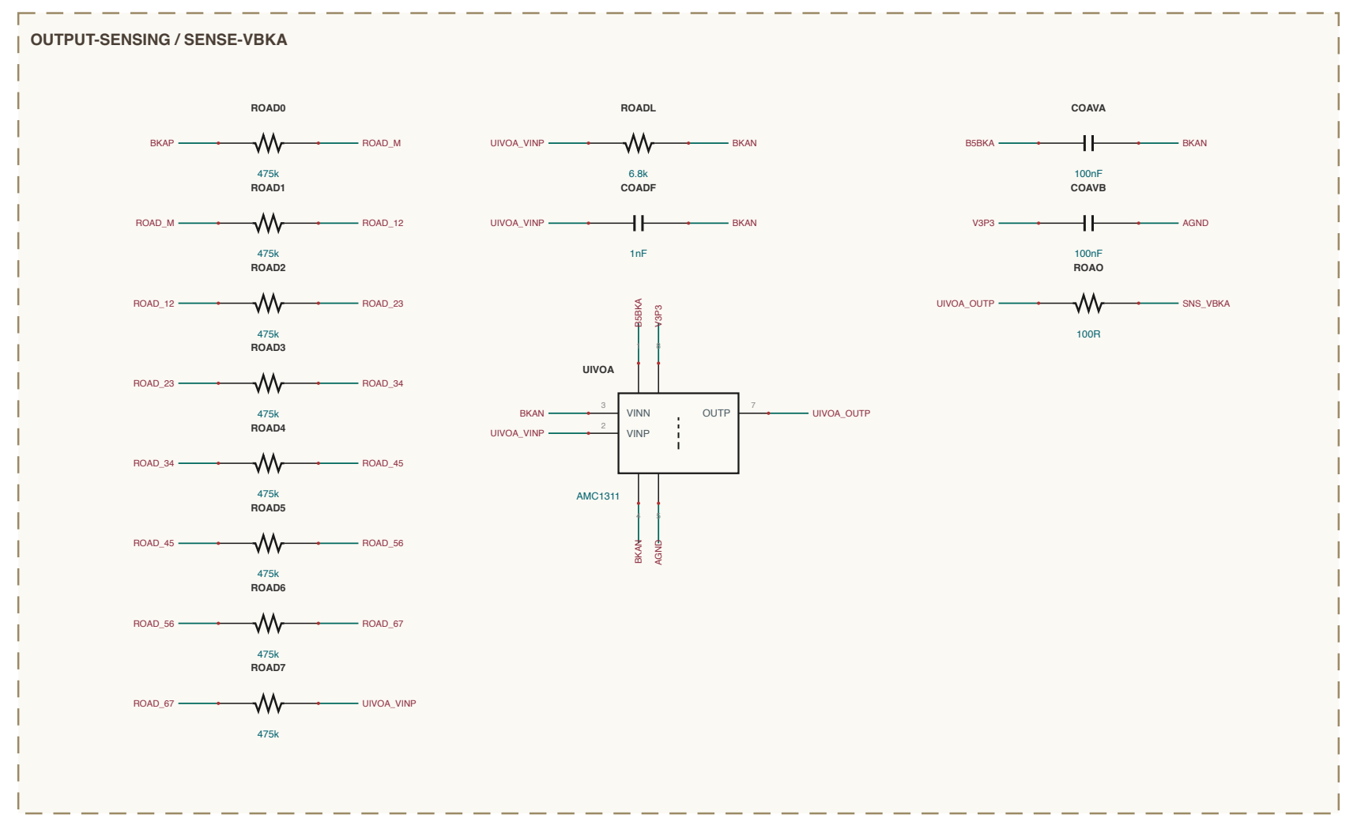

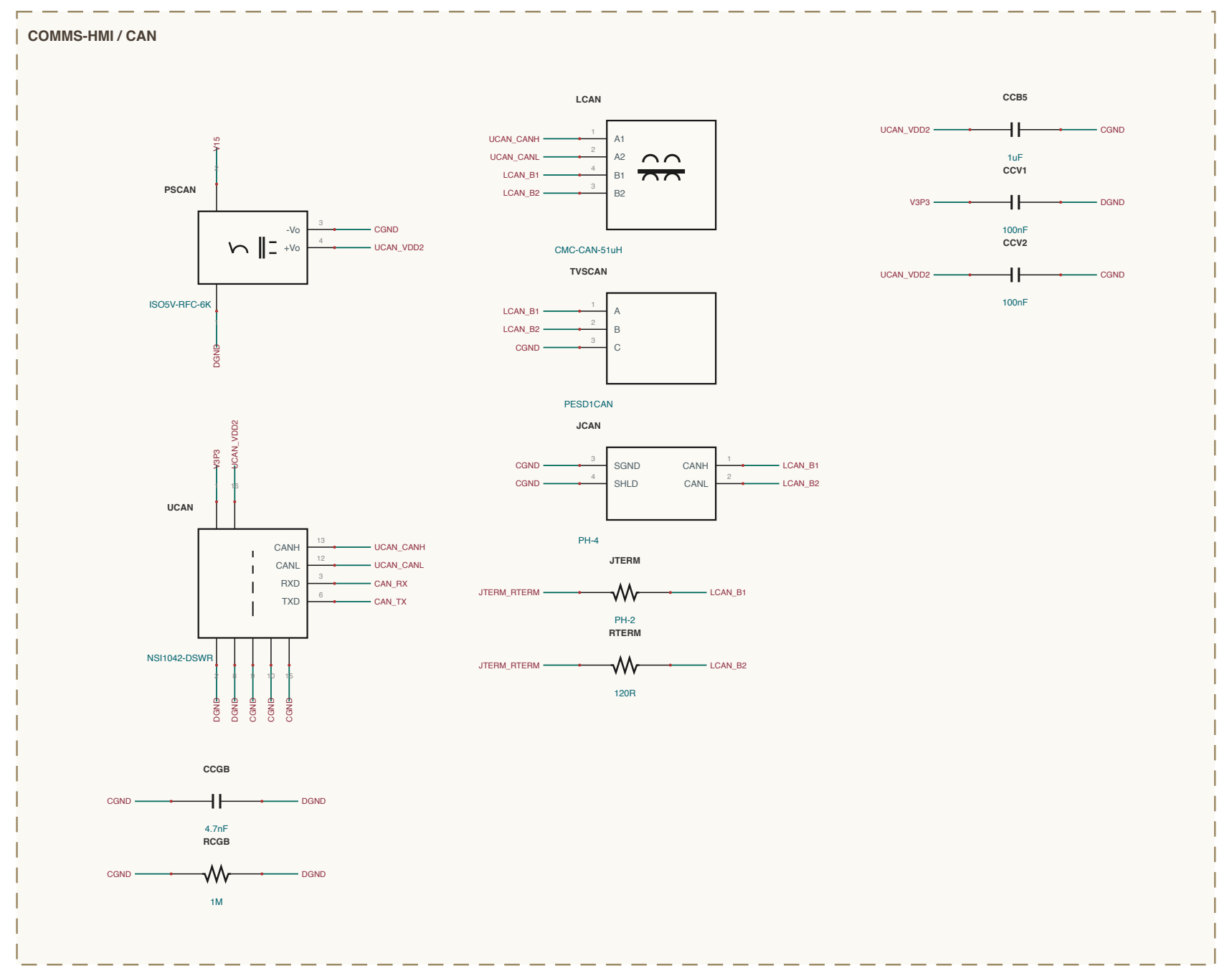

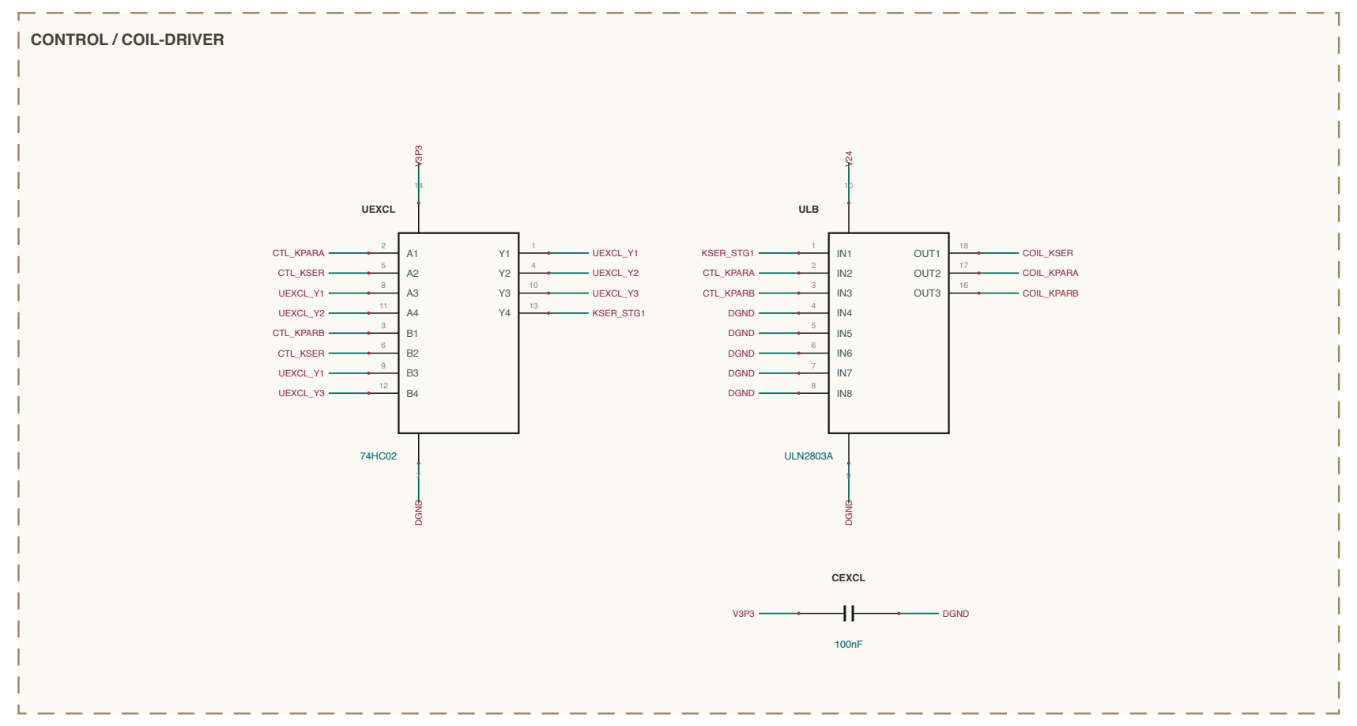

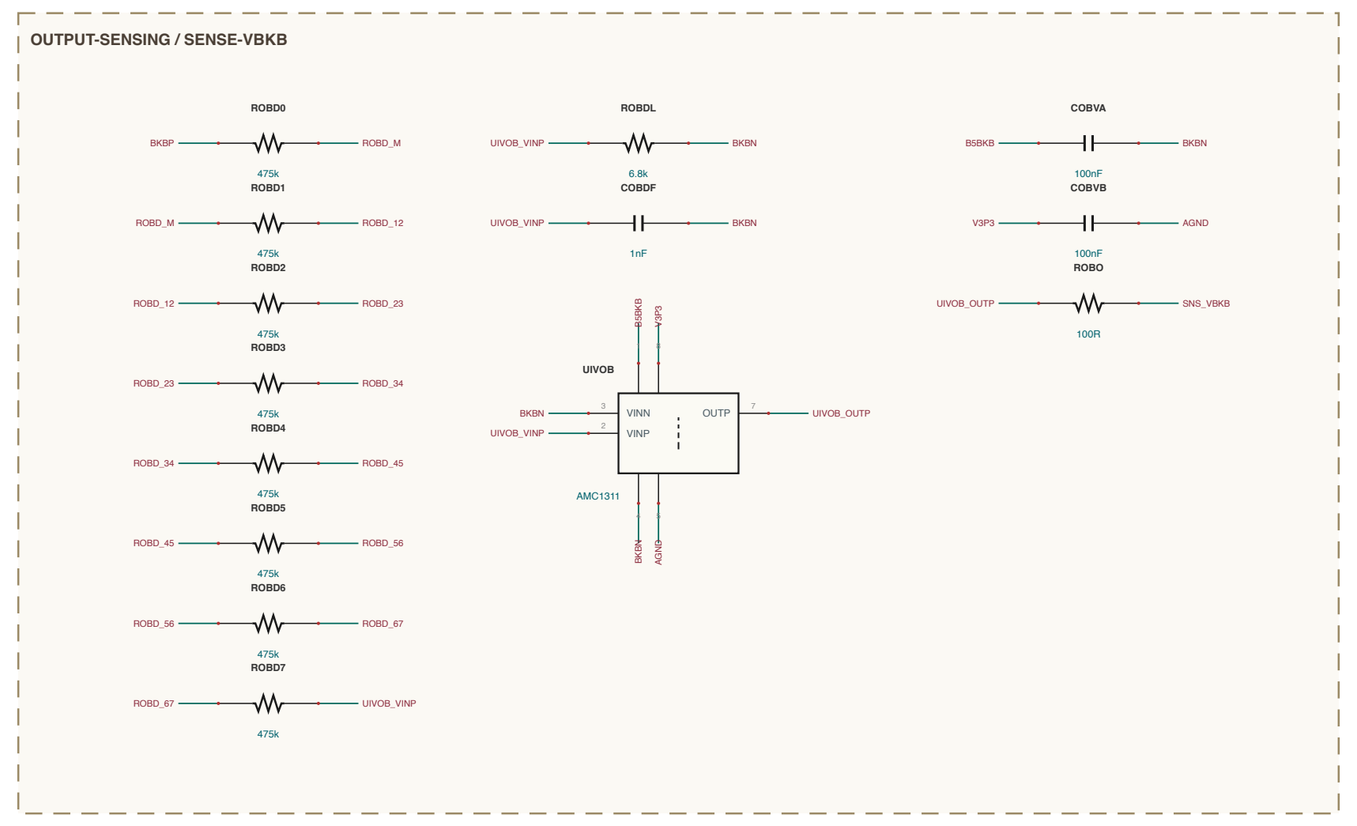

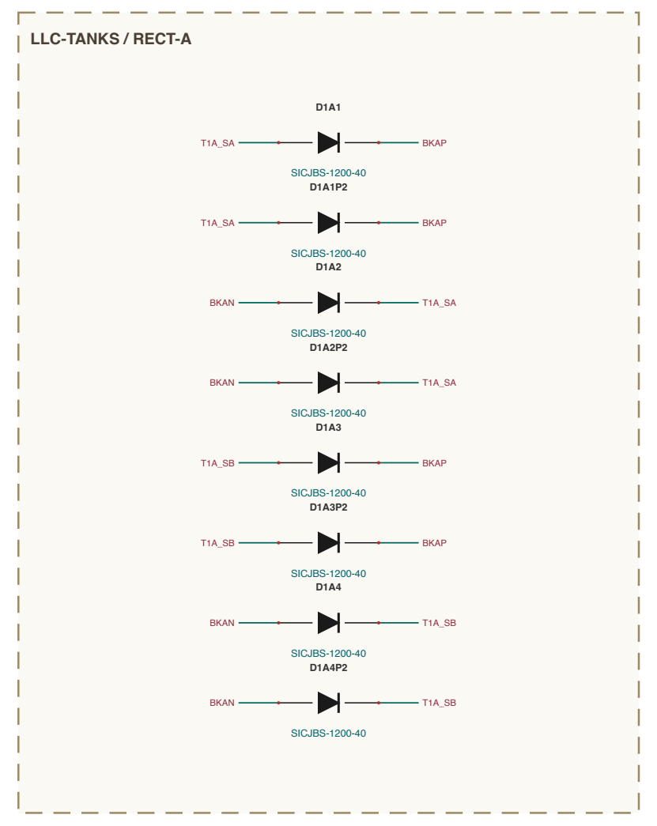

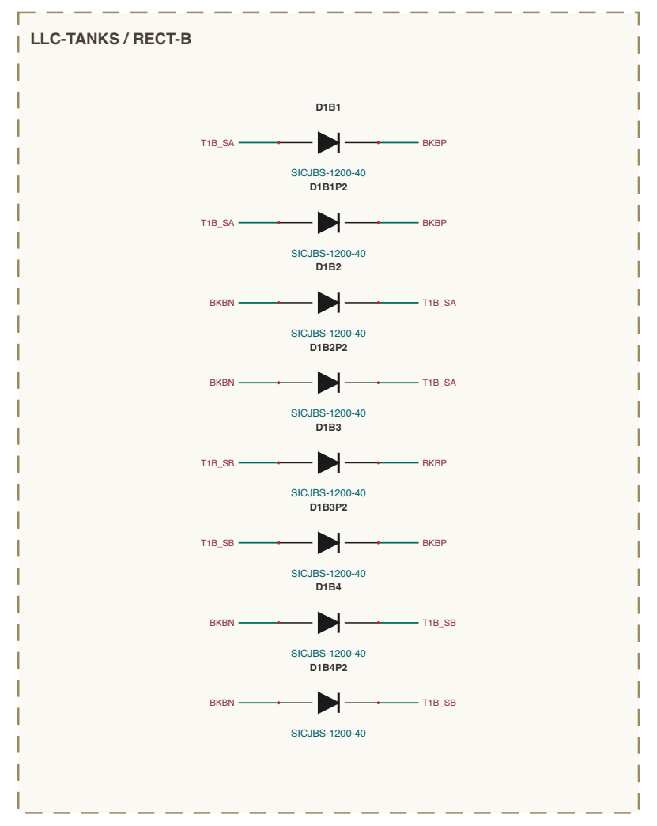

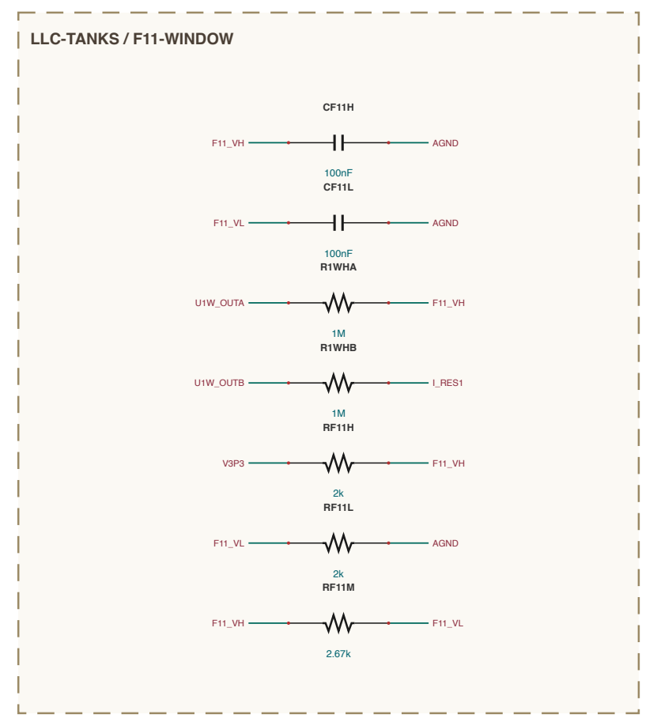

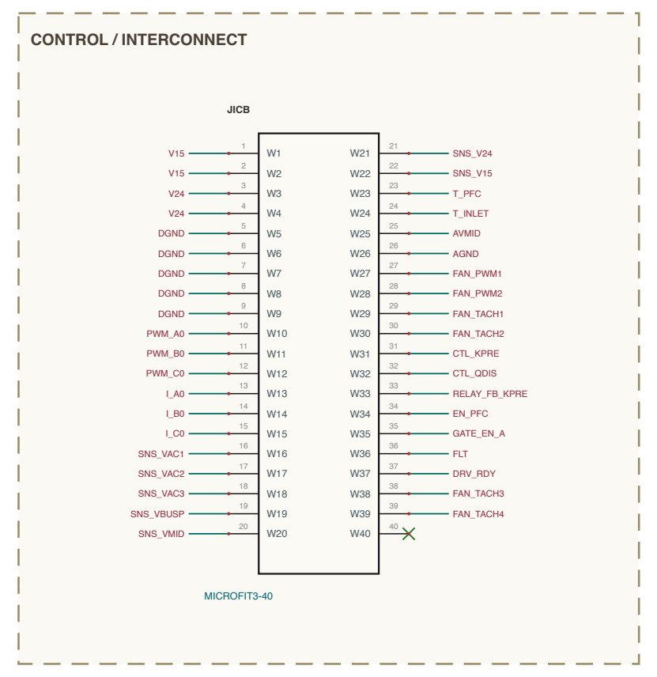

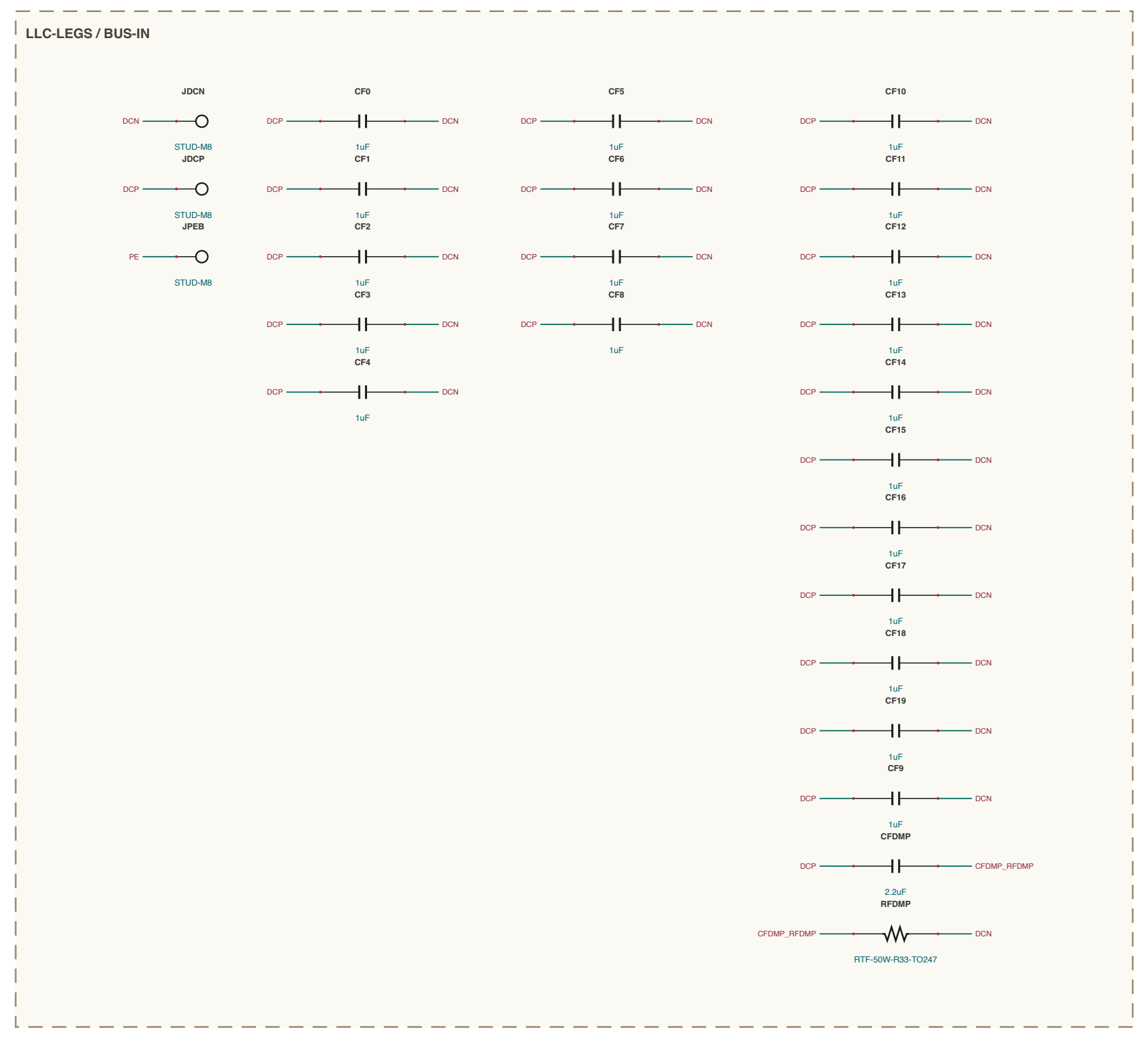

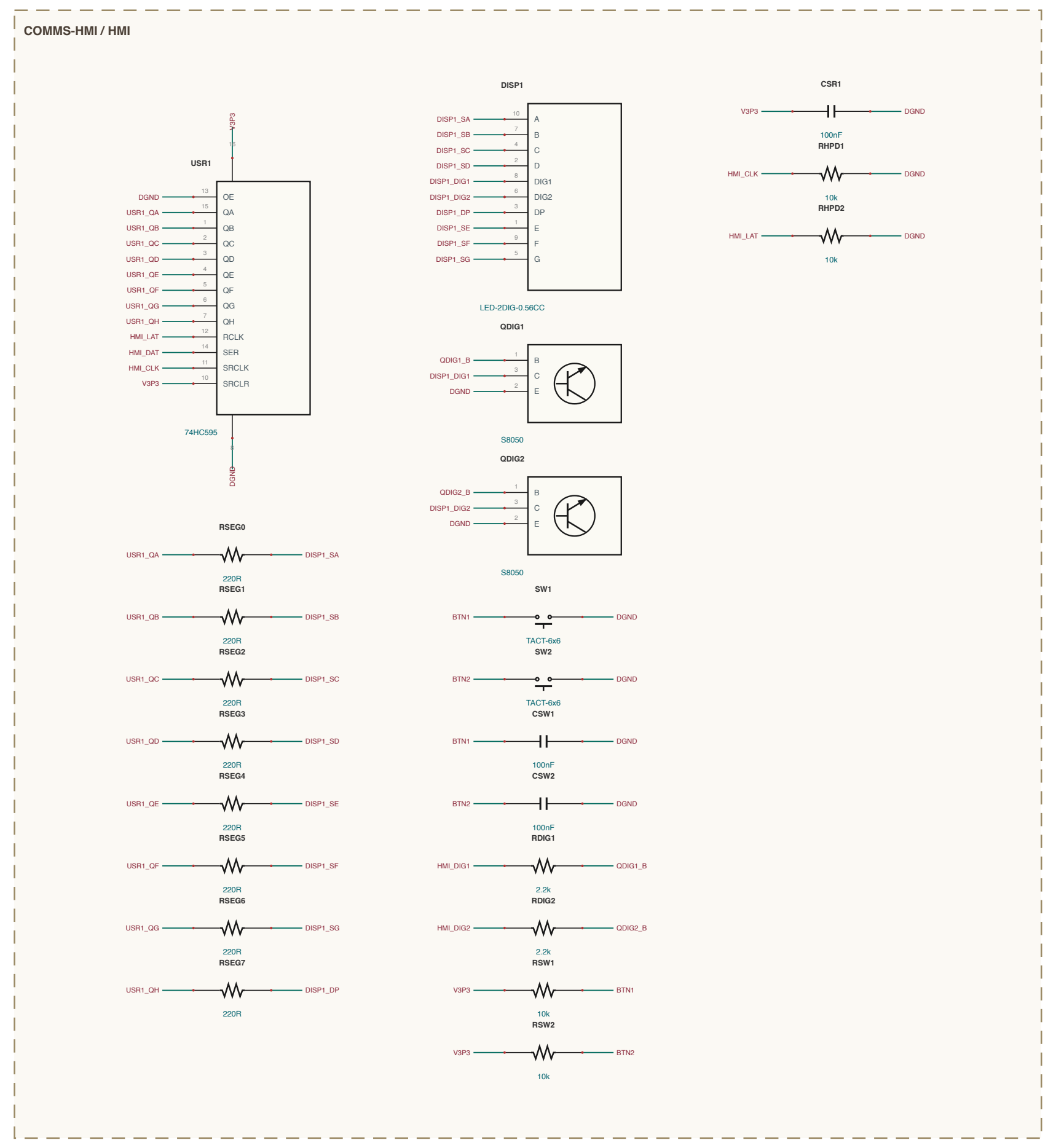

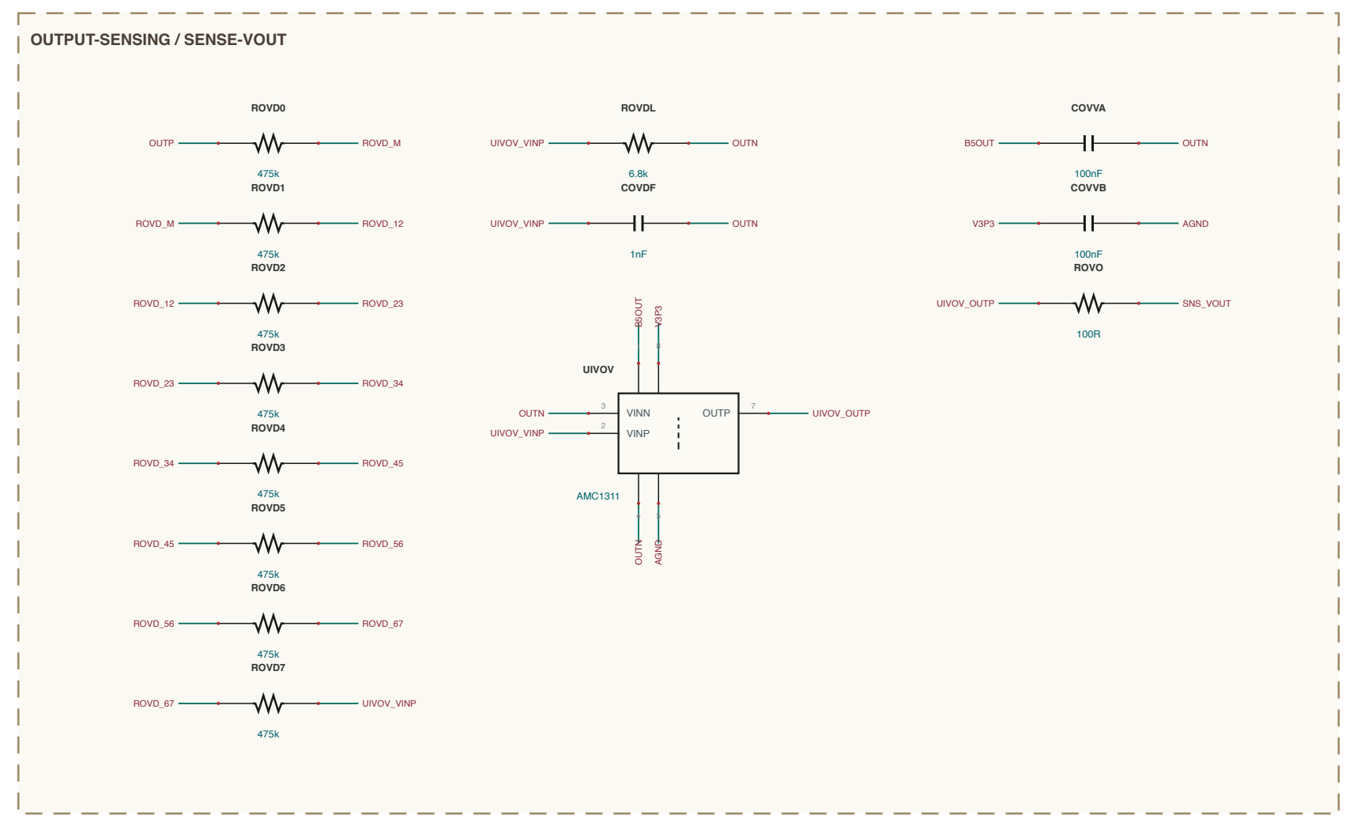

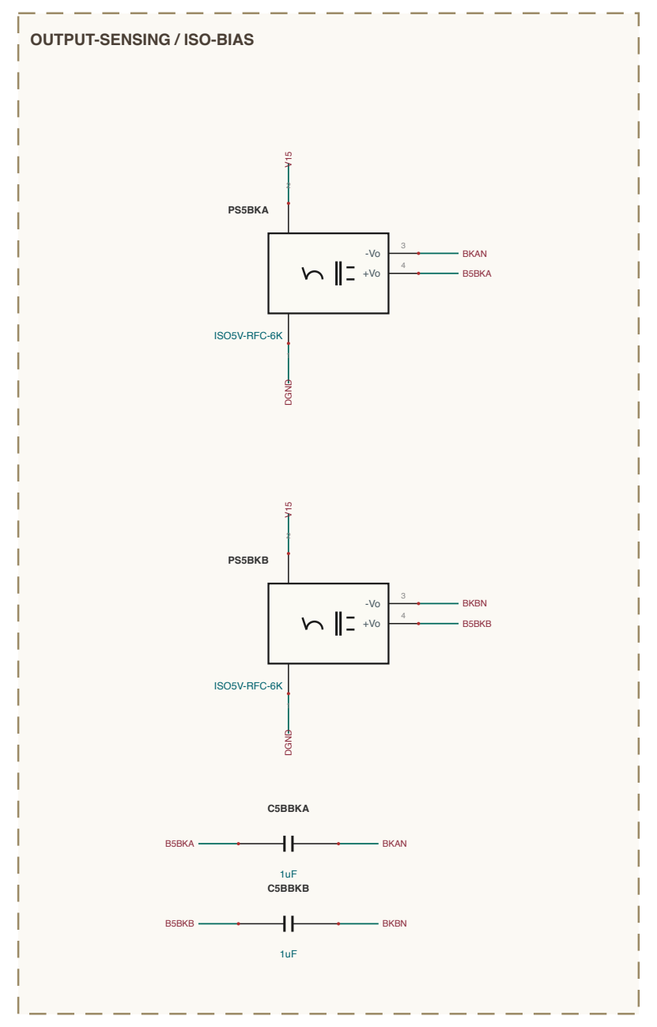

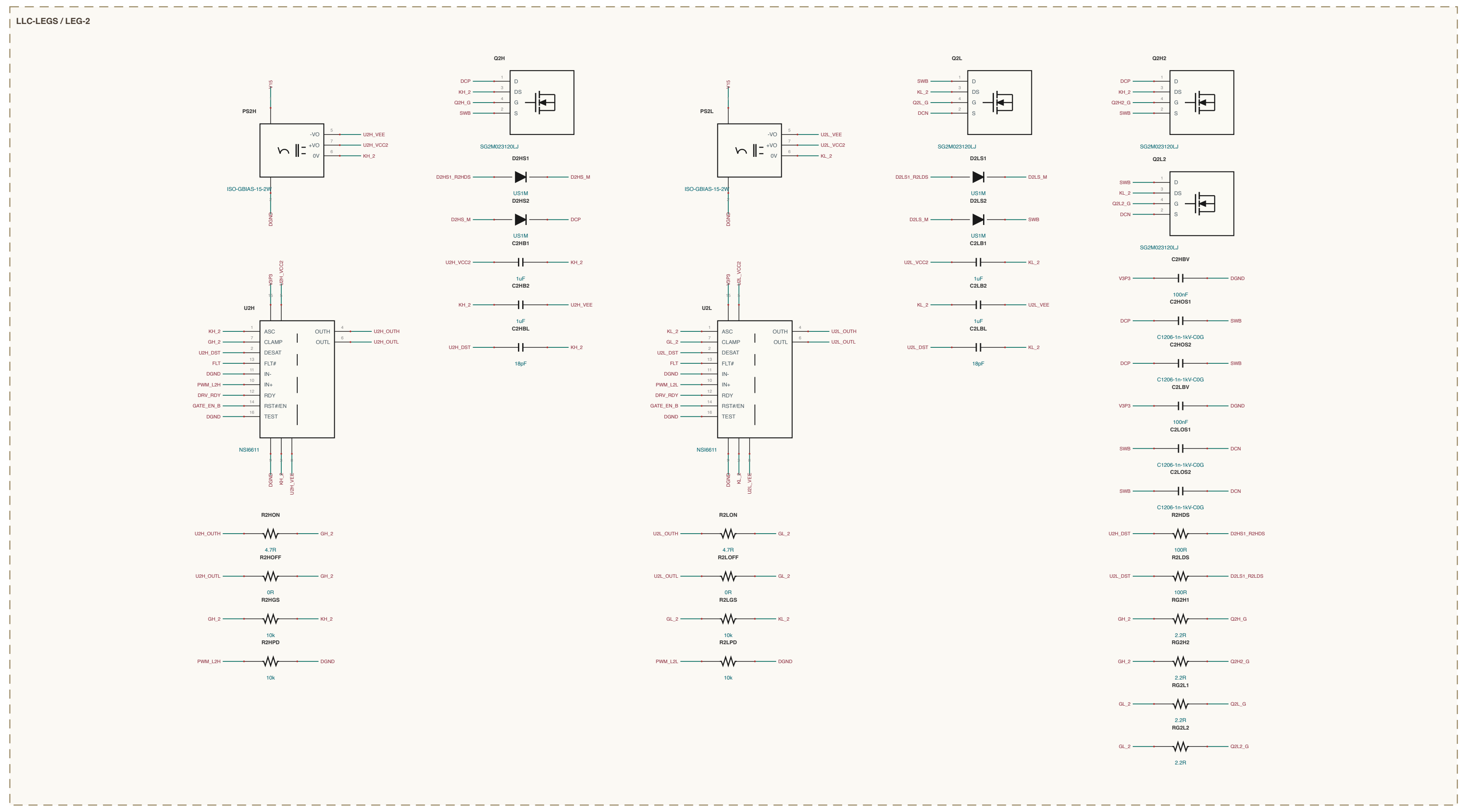

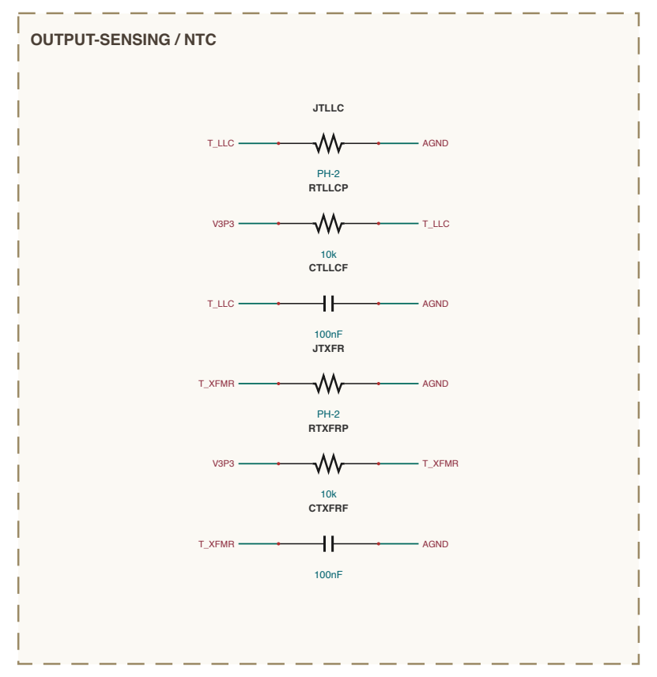

NET NAMING

internal junctions are named for what they join

U<ref>_<PIN> node at that IC pin e.g. UIVOA_VINP

R<stem>_M midpoint of a series pair e.g. RBALTA_M

R<stem>_<nm> tap between R<stem>n/m e.g. RV1D_01

all others are explicit design nets

MAGNETICS CONSTRUCTION

identity per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

D3 T1A/T1B (rev D): 2 cells, primaries in series -> n 2; each 3x E70/33/32 (3-set former), 4:4ll4, pri litz 3536x0.071 + sec 2x foil 0.08x28 halves, Lm 17.8uH/cell +/-7%, Bpk 159mT, plate/web-BONDED

D2 L1R (rev F): external Lr 3.20uH +/-3%, 2x E70/33/32, N=5, litz 12000x0.05, distributed gap, bonded

total Lr = D2 + 2x cell leakage (acceptance +/-30%) + 0.1uH loop = +/-5% simulated; Cr at TANK

bond loss screened by the EOL bonded thermal soak (T_XFMR NTC rise at fixed load)

SHEET INDEX

50 kW air DC-DC (upper) - 2 of 3

OUTPUT-SENSING - 6 sections

BANKS-SP - 4 sections

LLC-TANKS - 4 sections

CONTROL - 3 sections

LLC-LEGS - 3 sections

COMMS-HMI - 2 sections

rev D.4 - 22 sections - 316 components

DC-Modules 50 kW air — DC-DC board (full-bridge LLC)

DC-Modules 50 kW air - board DC-DC (upper), sheet 2 of 3

2026-09-06 · rev D.4